EDWARD VARVAK

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PROFILE

Analytical problem solver and machine learning practitioner with a Ph.D.-level mathematical background. Expertise in deconstructing complex problems and developing cutting-edge data-driven solutions. Proven skills in Python, computer vision, and deep learning, demonstrated through successful industry internships. Excellent communicator skilled at translating technical findings for diverse audiences.

WORK EXPERIENCE

Innovation Intern at PTC, *Summer 2021*
Researched and prototyped applications for state-of-the-art deep learning models for client use cases, focusing on computer vision and dimensionality reduction.

- Investigated and benchmarked automated object detection models to identify optimal architectures for resource-constrained environments.
- Co-developed an algorithm for estimating process cycle times from video footage, reducing the need for manual video analysis.

Machine Learning Intern at CIMCON, *Summer 2019*
Developed and deployed a real-time object-detection prototype on the NVIDIA Jetson Nano, demonstrating a proof-of-concept for a new product line.

- Engineered a gun detection model using TensorFlow and Keras for real-time video analysis.
- Evaluated low-power hardware, optimizing the model for continuous edge deployment.
- Presented technical results and project progress to R&D, sales teams, and senior leadership.

Computer Vision Software Intern at Eonite, *Summer 2017*
Contributed to an R&D team developing software for 3D mapping and localization in indoor environments.

- Improved camera-calibration tooling to reduce localization error and increase mapping accuracy.
- Implemented a C++ bundle-adjustment module to improve frame-to-frame localization by minimizing reprojection error.

EDUCATION

Doctor of Philosophy (Ph.D.) in Mathematics
University of Chicago — *Sep 2026–May 2027 (transfer)*
University of Illinois Chicago — *Aug 2022–Sep 2026*

- Repeat recipient of Research Training Group Fellowship, 4.0/4.0 GPA.
- Produced two first-author publications, invited speaker at numerous seminars and conferences.

Bachelor of Science (B.S.) in Applied Mathematics and Statistics
Stony Brook University — *Sep 2018–May 2022*

- Honors Track in Mathematics, Cum Laude, 3.6/4.0 GPA.
- Recipient of the Kuga-Sah Award for Excellence in Mathematics.

TECHNICAL SKILLS

Programming: Python (proficient), C++, C
ML Frameworks: PyTorch, Scikit-learn, TensorFlow
Tools and Hardware: Git, Debian Linux, Docker, NVIDIA Jetson
Core Competencies: Mathematical Modeling, Data Analysis, Problem Solving, Technical Presentations